

Background

- **No Limit Hold-em Poker (NLHE)**
 - Most popular version of Poker
 - Each player is dealt two private hole cards and five community cards are gradually revealed
 - Players decide on how many chips they want to put into the pot across four betting rounds
 - Win if everyone else folds or by having the best five-card hand



- Due to the private hole cards, poker is an **incomplete information** game

Purpose

- Poker solvers are a relatively recent development
 - They solve for the **Game Theory Optimal (GTO)** strategy, which is unexploitable and unbeatable
- Most high-quality poker solvers (like PioSOLVER and GTO Wizard) are commercial and expensive, despite using the same algorithms already published in academic literature

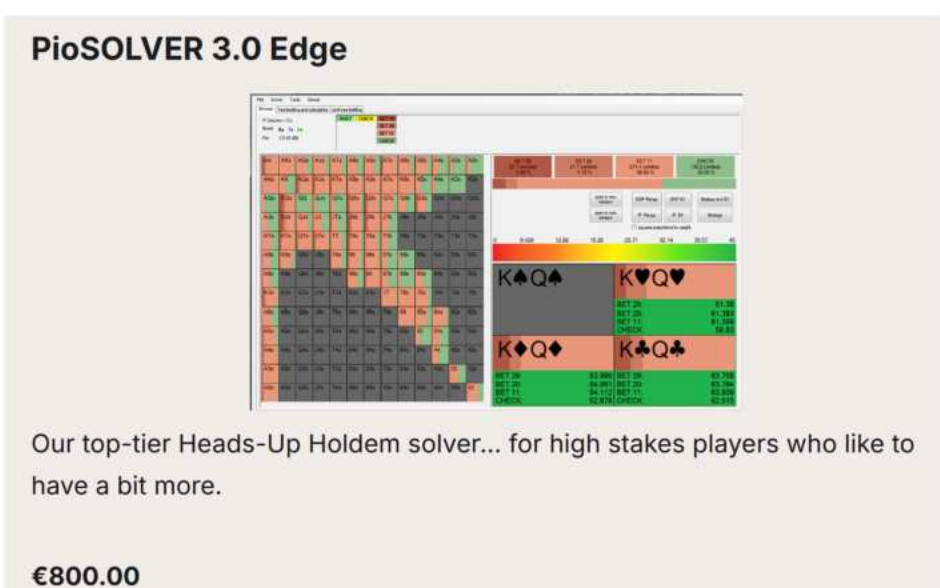


Figure 2: Screenshot of PioSOLVER 3.0 Edge taken by finalist from <https://piosolver.com/products>.

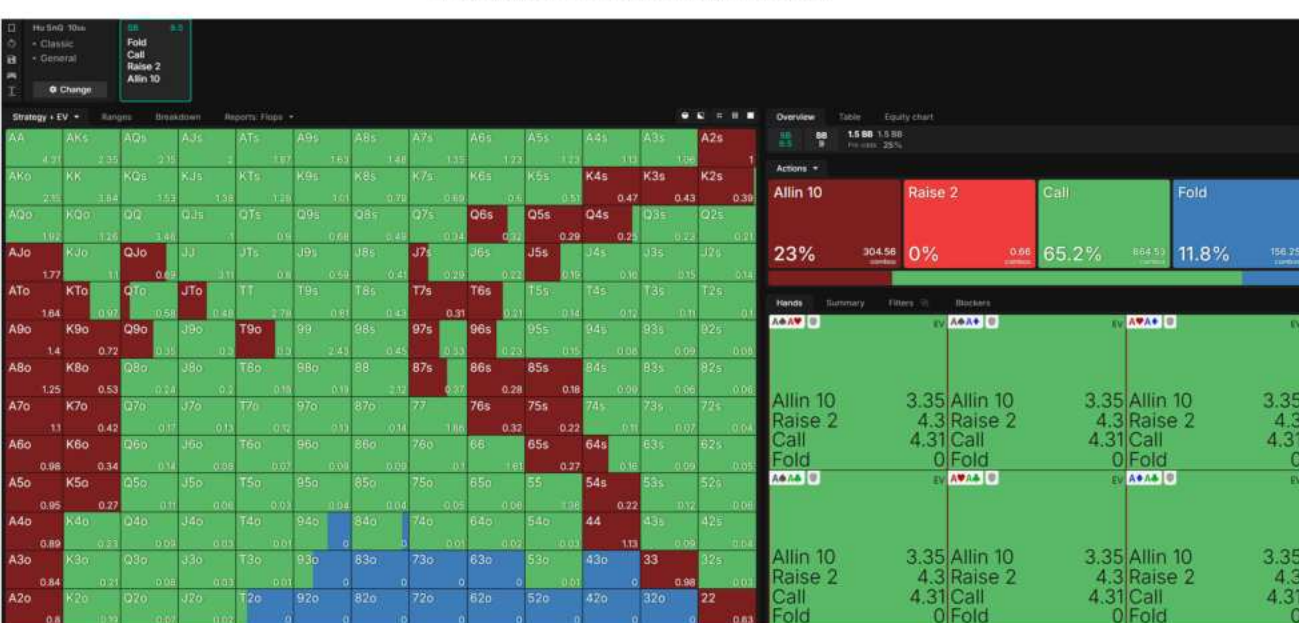


Figure 3: Screenshot of GTO Wizard Results UI taken by finalist from <https://app.gtowizard.com/solutions>.

- **Goal:** create an **open-source postflop NLHE solver** that solves for GTO

Nutsolver: Computing Optimal Poker Strategies

Solver Infrastructure

Game Tree

- NLHE can be represented as a **game tree**
- Each **node** contains information on the current gamestate and a player's strategy in that spot
- NLHE is too complex to fully solve, so the gametree is cut with abstractions (limitations on bet sizing)
- At each **terminal node**, a player wins the pot; this is the **expected value (EV)** at that node
- From there, the solver works backward to determine the EV of previous nodes

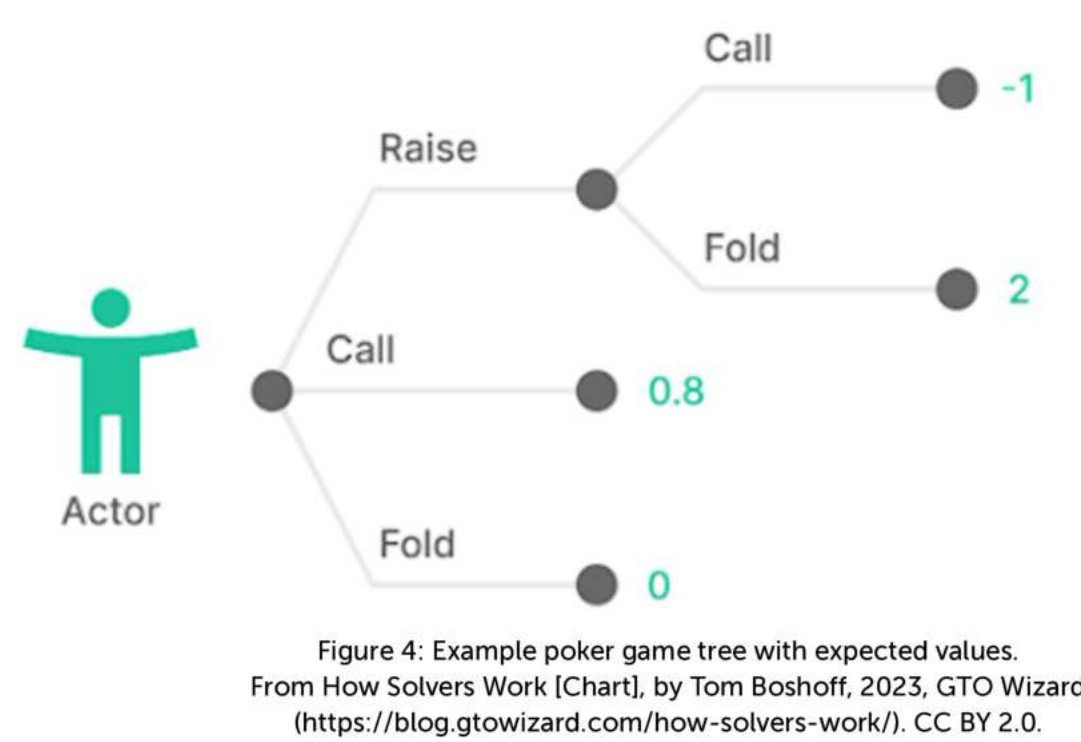


Figure 4: Example poker game tree with expected values. From How Solvers Work [Chart], by Tom Boshoff, 2023, GTO Wizard (<https://blog.gtowizard.com/how-solvers-work/>). CC BY 2.0.

Solving Algorithm

- **Regret:** how much the solver regrets not taking a different action based on EV
- Run two agents against each other with the goal of **minimizing regret**
 - Cannot take the highest EV strategy only or else they'll get stuck in a loop
- **Counterfactual Regret Minimization (CFR):** adjusts the current strategy by taking actions at a higher frequency the greater its positive regret is
- After many iterations, **Nash Equilibrium** will be reached where neither player can exploit the other player's strategy (GTO)

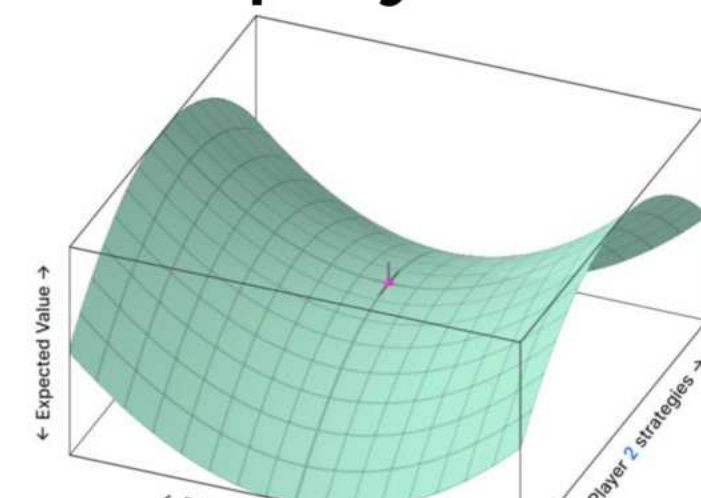


Figure 5: Graph of poker payoff space. From How Solvers Work [Chart], by Tom Boshoff, 2023, GTO Wizard (<https://blog.gtowizard.com/how-solvers-work/>). CC BY 2.0.

Software Design

- Currently, the solver can calculate equilibrium from any spot starting with all five community cards dealt
- Solver built in **C++** for fast runtime, GUI built in **Python**
- Player ranges, board cards, pot and effective stack size, and bet/raise sizes are defined in a .txt file and parsed by the solver
- Efficient hand evaluation done with **Cactus Kev's algorithm**
- **Chance-sampling Monte Carlo CFR** used for faster runtime
 - Randomly samples the cards dealt to each player to avoid iterating over the entire game tree
- The solver generates an output .txt results file, which is then parsed by a python script
- Data is converted into an interactive GUI with hand charts to visualize the results using **Tkinter** and **Matplotlib**

```
1 OOP: AA KK QQ JJ AKs AKo AQs AQo
2 IP: AA KK QQ JJ AKs AKo AQs AQo
3 BOARD: 2s 4d 2c 3s 4s
4 POT: 10
5 STACK: 100
6 OOP_BET_SIZES: 25 100
7 OOP_RAISE_SIZES: 50 100
8 IP_BET_SIZES: 25 50 75 100
9 IP_RAISE_SIZES: 25 50 75 100
```

Figure 6: Game configuration file. Screenshot of code created by finalist, 2026.

```
# =====
# Root Node
# =====

NODE root
ACTIONS: X, B50

HAND AcAd:
X 1.00
B50 0.00
```

Figure 7: Solver results output file. Screenshot of code created by finalist, 2026.

Results

Output from the game config in Figure 6:

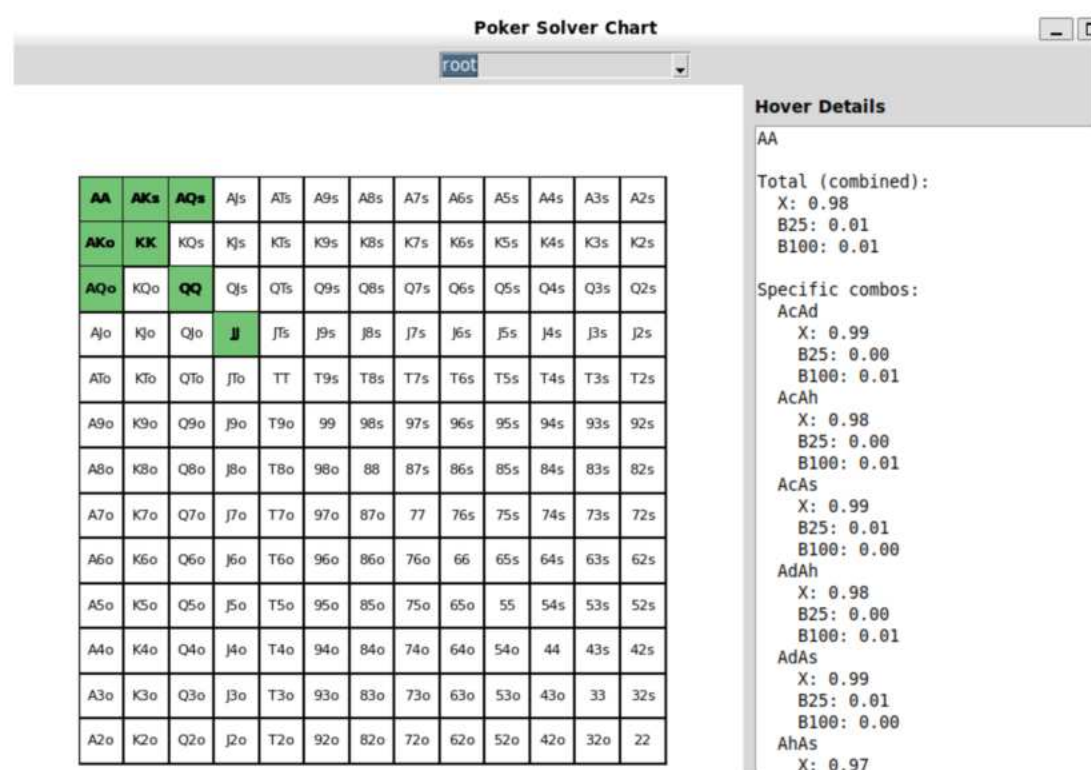


Figure 8: Solver results UI at root node. Screenshot of results UI created by finalist, 2026.

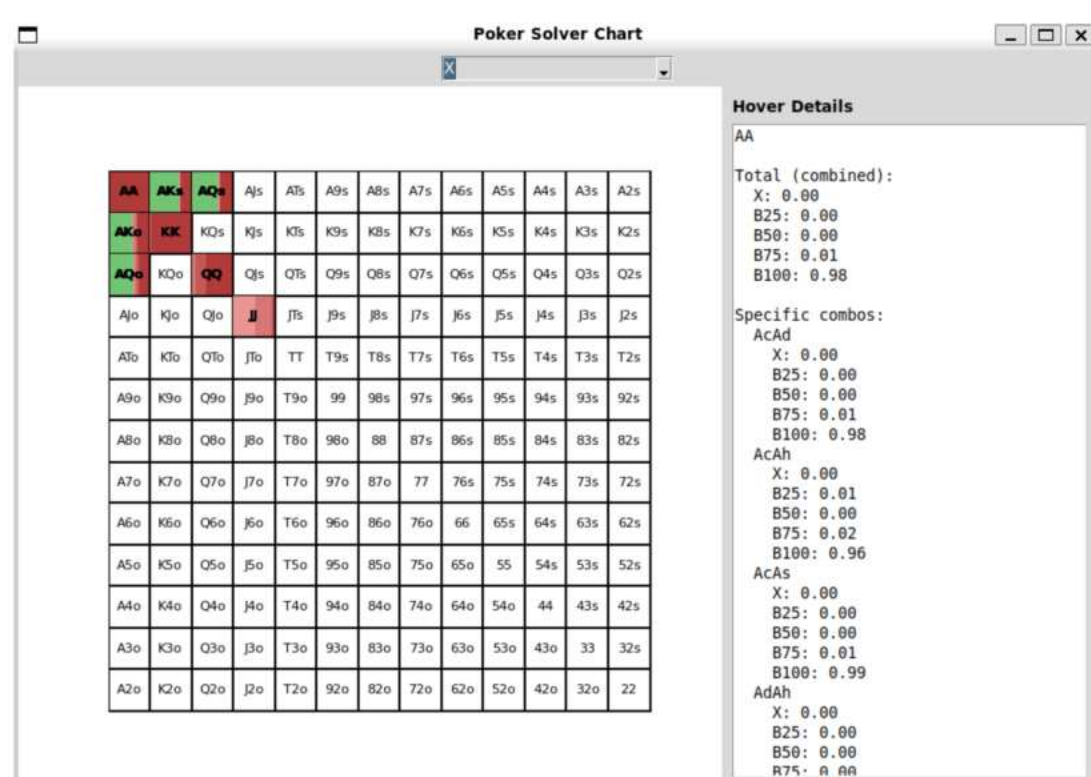


Figure 9: Solver results UI at X node. Screenshot of results UI created by finalist, 2026.

- These results show the solver's calculated equilibrium
- A bit of solver noise, but these results appear to be accurate based on principles of GTO play
- Solver is limited to computing equilibrium with all five community cards dealt; the next goal is to be able to solve from earlier betting rounds

Conclusion

- Successfully developed a solver that can calculate GTO from spots with all five community cards dealt
- Can be used to study or analyze hands
- Future goals: solving from earlier betting rounds and implementing other CFR algorithms and features

References

Boshoff, T. (2023, January 23). How Solvers Work. GTO Wizard <https://blog.gtowizard.com/how-solvers-work/>

Lanctot, M., Waugh, K., Zinkevich, M., & Bowling, M. (n.d.). Monte Carlo Sampling for Regret Minimization in Extensive Games. <https://mlanctot.info/files/papers/nips09mccfr.pdf>

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